

Section 1: What is a Use Case Diagram?

A use case diagram is a simple visual representation that demonstrates how users interact with a system. It describes what actions can be performed and who performs them. The focus is not on how the system is built internally but on what the system does from a user's point of view. A use case diagram helps identify the main features of a system and the interactions between different types of users and the system itself. It is typically created early in a project to capture the overall functionality in a clear and concise manner.

Section 2: Components of a Use Case Diagram:

A use case diagram includes three main components: actors, use cases, and relationships. Actors represent people, organizations, or external systems that interact with the application. Use cases represent the actions or functions the system can perform, such as placing an order, logging in, or making a payment. Relationships are the links that connect actors with their use cases, showing who implements which actions. There are also special types of relationships, such as "include" and "extend," which show dependencies or optional behaviors between different use cases.

Section 3: Purpose of a Use Case Diagram:

The core purpose of a use case diagram is to provide a clear understanding of how a system behaves from the user's perspective. It helps communicate the system's functionality between developers, designers, and stakeholders. By focusing on what users can do, it ensures that all important requirements are captured early in the design phase. Use case diagrams are useful for planning, identifying missing features, and confirming that the system will meet user expectations before the development process begins.